

GroupMate — Evaluation Dataset & Results

Building AI Application Challenge 2026 | Rana Hamad | Day 7 | 2026-09-11

Methodology

12 sample project descriptions (docs/test-project-descriptions.md) were run against the live OpenAI API (gpt-4o) on 2026-09-11, exercising both AI calls end-to-end: `generate_tags()` and `ai_distribute_tasks()`. Samples span software, research, creative, and event-planning projects, with group sizes from 1 to 6 members, including members with no strong suits selected. Full reproduction script: `scripts/run_eval.py`.

Metrics

Tag validity: `generate_tags()` returns 6-10 relevant, project-specific tags as valid JSON. **Task/name validity:** `ai_distribute_tasks()` returns valid JSON where every assigned_to/shared_with name matches a real member exactly (0 unmatched names = pass). **Fairness:** spread (max-min) and standard deviation of estimated hours across members — lower is fairer.

Summary Results

Metric	Result
Samples run	12 / 12 completed, 0 failures
Tag JSON validity	12 / 12 (100%) — 6-10 relevant tags every time
Name-matching validity	12 / 12 (100%) — 0 AI-assigned names failed to match a real member
Average hour spread (11 multi-member samples)	10.3h (min 2.0h, max 25.3h)

Per-Sample Results

#	Project (excerpt)	Members	Tags	Tasks	Hours per member	Spread
1	Bill-splitting mobile app	3	10	6	Ali 25 / Sara 25 / Omar 20	5.0
2	Remote-work economic impact report	2	10	8	Layla 38.5 / Yousef 41.5	3.0
3	Sustainable fashion marketing campaign	4	10	5	Mona 20 / Fahad 15 / Nour 10 / Zaid 5	15.0
4	Full-stack e-commerce website	4	10	7	Hassan 38.8 / Reem 28.8 / Tariq 28.8 / Dana 15	23.8
5	Mental-health documentary film	2	10	4	Huda 13 / Karim 17	4.0
6	ML model for grade prediction	5	10	8	Salma 18 / Adel 16 / Rania 11.7 / Bilal 3.7 / Nabil 3.7	14.3
7	Charity fundraising event	3	10	3	Yasmin 12 / Khaled 10 / Lina 10	2.0
8	Discord scheduling bot	1	9	4	Marwan 30	N/A (1 member)
9	Food-delivery app UX research	3	10	7	Farah 19.7 / Ibrahim 18.3 / Noor 15	4.7
10	Climate-change board game	6	10	5	Aisha 15 / Waleed 15 / Huda 15 / Sami 3.8 / Talal 3.8 / Rasha 3.8	11.2
11	Internship-posting scraper tool	2	10	8	Ahmad 29 / Dalia 24	5.0

#	Project (excerpt)	Members	Tags	Tasks	Hours per member	Spread
12	Entrepreneurship podcast series	4	10	15	Jana 15 / Sultan 10.7 / Maya 36 / Fadi 17.3	25.3

Analysis & Findings

Strengths: JSON reliability was perfect across all 12 runs — no retries were triggered, and every task's assigned_to/shared_with names resolved to a real member with zero mismatches. Simpler, single-deliverable projects (samples 2, 5, 7, 9, 11) produced tight, fair hour splits (2.0-5.0h spread).

Weakness found: larger or more heterogeneous projects (sample 4: e-commerce site with distinct frontend/backend/payments/QA roles; sample 12: a 6-member-worth podcast series with 15 generated tasks) produced noticeably wider spreads (23.8h and 25.3h). This suggests the model weighs skill-tag match more heavily than strict hour-balancing when a project has clearly specialized roles — reasonable in principle (a payments specialist should get payment tasks), but worth tuning the prompt to more explicitly trade off skill-fit against hour-balance for larger projects.

Next step: not implemented in this challenge window — would adjust the ai_distribute_tasks() system prompt to cap the maximum allowed hour spread as a percentage of total project hours, re-running the model if the first attempt exceeds it.